

Monthly model algorithm

Local-currency DeFi credit-loss stress test

Inputs. 42 joint monthly-average FX/crypto returns; assumed CR grid and weights; a lagged rate rule; threshold $\theta = 1.5$; delay $d \in \{0, 1, 2\}$; haircut h_0 , congestion slope γ , cap 0.30; penalty 0.13; reserve $k = 0.10$.

Initialization. Sample non-circular three-month joint blocks to obtain 14 months; truncate the final block. Set FX, crypto and LCU debt factors to one. Set month-1 rate to 20%. Set every cohort active, with no scheduled sale; initialize USD settlement capital to k per initial portfolio USD. One matching LCU funding principal is outstanding per loan.

For each path and month $t = 1, \dots, 14$:

1. **Observe prices.** Update FX factor $X_t = X_{t-1}(1 + \delta_t)$ and crypto factor $Z_t = Z_{t-1}(1 + g_t)$.
2. **Set rate.** Apply the specified rule to returns dated strictly before t ; use expanding history during initialization. Apply its floor, cap and adjustment limit.
3. **Accrue debt.** For unpaid loans, update $U_t = U_{t-1}(1 + r_t/12)$ and USD debt $D_t = U_t/X_t$.
4. **Value collateral.** For CR cohort j , collateral is $C_{jt} = c_j Z_t$ and $CR_{jt} = C_{jt}/D_t$.
5. **Register breach.** Through month 12, queue the first strict breach $CR_{jt} < 1.5$. Record breach month τ_j . The decision is irrevocable.
6. **Apply delay.** Schedule sale for $e_j = \tau_j + d$. Debt keeps accruing. A recovery of CR does not cancel a queued sale.
7. **Execute due sales.** Sum due initial-exposure weights q_t and set $h_t = \min(0.30, h_0 + \gamma q_t)$. Sell all due collateral; net proceeds are $a_j = (1 - h_t)C_{jt}$.
8. **Allocate proceeds.** Recover $R_j = \min(D_t, a_j)$; record $L_j = D_t - R_j$; collect penalty $\pi_j = \min(0.13D_t, [a_j - D_t]_+)$; return $V_j = [a_j - D_t - \pi_j]_+$. Mark the position settled.
9. **Update funding and outcomes.** Retire matching funding principal at $1/X_t$ USD per initial USD. Add weighted $R_j + \pi_j - 1/X_t$ to capital. Record financing effect $1 - R_j - \pi_j - h_t C_{jt}$. At month 12, unbreached loans repay at par, retire funding and retain collateral. Only scheduled sales remain for months 13–14.

Aggregate. Separately report breach, executed, pending and undercollateralized exposure at month 12; all close-out credit losses; survivor/liquidated/ex ante financing effects; and gross credit-budget and matched-funding gap indicators. Estimate CVaR₉₉ from the worst 1% of path losses. Never label the funding-gap statistic a peg-failure forecast.